STELLAR: Similarity-Based Satellite Federated Learning for Malicious Traffic Recognition

Yubo Li, Li Zhang, Kai Li, and Haoru Su

***Abstract*—Resource constraints and data heterogeneity pose significant hurdles for malicious traffic detection in satellite networks. To address this, we propose STELLAR, a similarity-based federated learning framework tailored for efficient space computing. STELLAR introduces a multi-dimensional similarity metric to dynamically select representative nodes, effectively eliminating computational redundancy. Furthermore, it ensures system robustness and trust through a time-window asynchronous protocol with staleness compensation and a lightweight proxy-based authentication scheme. Evaluations demonstrate that STELLAR outperforms specialized baselines, reducing network-wide energy consumption by 77%–80% while achieving 99.72% detection accuracy in heterogeneous Non-IID environments. These results validate STELLAR as a sustainable and robust solution for distributed security in resource-constrained satellite networks.**

***Index Terms*—Federated learning, LEO satellite networks, intrusion detection, multi-dimensional similarity, asynchronous aggregation, lightweight authentication.**

I. Introduction

Satellite networks, operating in open space environments, are inherently vulnerable to malicious attacks. Satellite nodes are characterized by strictly limited computing power, storage capacity, and energy supply, coupled with low-bandwidth and unstable inter-satellite links. Furthermore, satellite traffic differs sharply from terrestrial Internet patterns, dominated by specialized services such as remote sensing and meteorological data. Attackers often exploit these characteristics to launch energy-targeted strikes (e.g., during low-energy eclipse periods). These constraints hinder the deployment of computationally intensive terrestrial detection models, such as complex deep learning architectures, directly onto satellites.

Deep learning methods have adapted neural networks for satellite environments with promising performance; however, they often fail to meet practical requirements due to resource bottlenecks [1]. A critical hurdle is the scarcity of open, authentic, and diverse satellite traffic datasets. Emerging applications and novel attack vectors lead to high rates of unknown traffic, rendering models trained on outdated or terrestrial datasets ineffective. Moreover, transmitting large-scale raw traffic to terrestrial ground stations for centralized training incurs prohibitive communication overhead and latency, raising significant concerns regarding privacy, data confidentiality, and timeliness. Training models on satellites can address the transmission overhead caused by dataset updates, but even lightweight continuous online learning imposes a severe strain on scarce space resources.

Federated Learning (FL) offers a paradigm shift by enabling data-localized collaborative training, which mitigates data scarcity and transmission overhead by aggregating distributed on-board insights, theoretically suiting capability-limited nodes. However, directly applying terrestrial FL frameworks to satellite networks faces unresolved challenges: data distribution heterogeneity (Non-IID), low communication efficiency, and security vulnerabilities. For instance, many algorithms require frequent transmission of large models to a central server [2], yet satellite links—characterized by high latency and dynamic topology—render such large-scale exchange inefficient. Although local training preserves privacy, it suffers from uneven dataset distributions and feature skew, leading to slow convergence or model divergence. Crucially, neighboring satellites often share similar coverage and mission tasks, leading to highly redundant training that incurs unnecessary energy expenditure. Additionally, traditional synchronous aggregation protocols are fragile; a few straggler nodes caused by link instability can stall the entire training process. Finally, while most frameworks focus on data privacy, they often overlook participant trustworthiness. In the open space environment, node hijacking is a tangible risk, as the inclusion of a single malicious node can poison the global model.

To address these challenges, we propose **STELLAR**: **S**imilarity-based sa**T**ellite f**E**derated **L**earning for ma**L**icious tr**A**ffic **R**ecognition. Tailored for resource-constrained satellite networks, STELLAR leverages collaborative federated learning to identify malicious traffic. It effectively mitigates single-node data limitations while significantly reducing communication and computational overhead. Representative nodes are selected via data similarity to reduce computational overhead, while multi-layer aggregation coordinated by an asynchronous protocol minimizes costs and ensures survivability against stragglers. It employs a lightweight authentication scheme utilizing cryptography to authenticate entities, form keys, and ensure participant trustworthiness and transmission security.

The main innovations and contributions of this paper are summarized as follows:

(1) **Multi-dimensional data similarity measurement for satellite robustness**: We propose a composite metric that fuses model parameters, convergence patterns, and prediction behaviors. This metric accurately character-

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izes intrinsic correlations within data distributions and functions as a semantic filter, ensuring robust grouping even in the presence of high transmission noise in satellite links.

(2) **Energy-efficient representative mechanism**: We introduce an adaptive grouping-based Representative Node mechanism to mitigate redundant computation and reduce network-wide energy consumption.

(3) **Asynchronous protocol for survivability under instability**: We design a time-window-based asynchronous aggregation protocol equipped with a staleness compensation mechanism. This design enables the system to effectively leverage late updates and maintain model convergence despite severe delays or interruptions, ensuring system survivability in harsh space environments.

(4) **Orbit-adaptive quality-aware hierarchical aggregation**: We develop an aggregation mechanism tailored to the dynamic orbital topology. By incorporating a *quality-aware* weighting strategy, it balances model accuracy with data scale, guaranteeing global model stability even when individual satellites provide low-quality updates due to resource fluctuations.

(5) **Lightweight trust via proxy-based authentication**: We devise a proxy-based satellite node authentication scheme to address strict security requirements and environmental constraints. It achieves secure node authentication and key negotiation with minimal latency and computational overhead, ensuring both framework security and operational efficiency.

II. Related Work

Deep learning approaches, such as BiLSTM [3], BiTCN+MHSA [4], and GAN+RF [5], have achieved breakthroughs in terrestrial malicious traffic detection through automatic feature extraction. However, these methods typically incur high computational complexity and require substantial resource consumption.

Due to the unique characteristics of satellite networks, terrestrial methods are ill-suited for direct deployment. While recent works have explored neural network optimization for satellite environments [6], [7], their simulation scenarios often oversimplify the dynamic topology of integrated satellite-terrestrial networks [6], or they exhibit excessive complexity that hinders on-board deployment [7].

Federated learning (FL) enables collaborative training while preserving data locality. Early satellite FL works [8], [9] proposed distributed intrusion detection systems but failed to address the Non-IID nature of satellite traffic, leading to performance degradation. Recent advances have incorporated anti-adversarial mechanisms [10], mesh architectures [11], and synthetic data generation [12]. Nevertheless, these methods face several limitations: (1) performance deterioration when the proportion of malicious participants is high [10]; (2) difficulties in assessing node trust [11]; and (3) risks of feature leakage combined with a neglect of orbital dynamics [12]. Critically, most existing approaches assume full participation or frequent synchronous updates, which imposes unbearable energy costs on satellites. STELLAR addresses this via similarity-based representative selection to significantly reduce energy consumption while ensuring convergence.

Clustering methods (e.g., CFL [13], FeSEM [14]) improve FL convergence on Non-IID data via personalized modeling. However, they fundamentally rely on *full client participation*—requiring every node to continuously perform local training for clustering—which is impractical for energy-starved satellite networks. Similarly, Hierarchical FL [15] enhances cloud-edge efficiency but depends on static edge servers and simple size-weighted averaging ($\frac{|\mathcal{D}^l|}{|\mathcal{D}|}w^l$). In contrast, STELLAR adapts to the dynamic orbital topology through quality-aware weighting and representative rotation.

Beyond efficient training, ensuring the integrity of the collaborative process is equally critical. In the domain of satellite security, Abdrabou et al. designed cross-orbital key agreement with link switching prediction, maintaining stable short-term delays in inter-satellite switching [16] while imposing heavy computational burdens on satellite payloads.

To address satellite computational and storage constraints, lightweight authentication is key. Symmetric encryption-based protocols have been extensively explored, such as the constellation maintenance schemes by Huang et al. [17] and the LEO identity authentication methods by Zhu et al. [18], [19], which utilize pre-computation to reduce overhead. Other lightweight approaches include non-interactive elliptic curve cryptography [20], hash-XOR hybrid frameworks [21], and proxy signature verification [22]. Building on these foundations, this paper designs an authentication scheme to further reduce overhead.

III. STELLAR: Similarity-Based Satellite Federated Learning Framework

A. Design Objectives and Problem Formulation

STELLAR aims to establish an effective malicious traffic detection system for resource-constrained satellite networks. To achieve this, we adopt Federated Learning (FL) as the training paradigm, which enables satellites to collaboratively train a detection model without centralizing raw data. Within the FL framework, STELLAR is designed to fulfill three objectives:

Objective 1: Energy Efficiency with Accuracy Preservation. Given the severe energy constraints of satellite platforms, the framework must minimize computational and communication resource consumption while maintaining detection accuracy:

$$\min E_{total}^{STELLAR}, \quad \text{s.t.} \quad \text{Accuracy} \geq \eta_{acc} \tag{1}$$

where $E_{total}^{STELLAR}$ denotes the total energy consumption, and η_{acc} is the target accuracy threshold.

This objective is addressed by the *similarity-based satellite selection strategy* (Section III-D), which groups satellites with similar data distributions and selects representative nodes for training, significantly reducing the number of active participants.

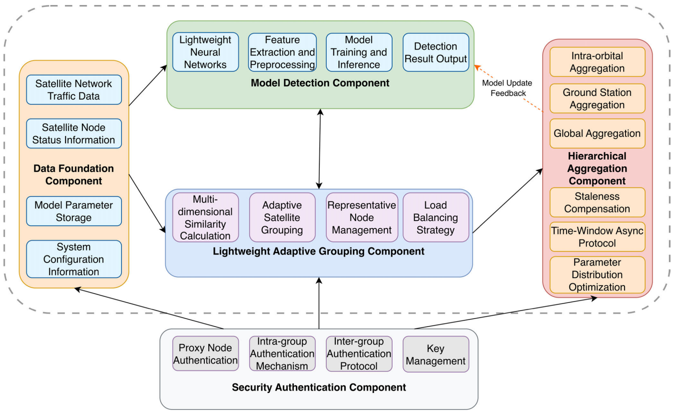


Fig. 1. Stellar framework architecture.

Objective 2: Convergence Guarantee. The global model must converge to an ϵ-stationary point despite partial participation and communication instability:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(w_t)\|^2 \leq \epsilon \tag{2}$$

where $F(\cdot)$ denotes the global loss function, w_t is the global model parameter at round t, and T represents the total number of training rounds.

This objective is addressed by the *quality-aware hierarchical aggregation mechanism* (Section III-E) and the *asynchronous aggregation protocol with staleness compensation* (Section III-F), which ensure stable convergence under heterogeneous data and intermittent links.

Objective 3: Security Guarantee. Operating in an open space environment, the framework must ensure: (a) only authenticated satellites participate in training; and (b) model parameters are protected during transmission.

This objective is addressed by the *lightweight LASS-PN authentication protocol* (Section III-G), which achieves secure node authentication and key negotiation with minimal computational overhead.

The Core Challenge. Objectives 1 and 2 are inherently conflicting. Standard FL requires all N satellites to train every round, achieving convergence but imposing prohibitive energy costs ($E_{total}^{FedAvg} = T \cdot N \cdot e_{act}$). The key question STELLAR addresses is: *What is the minimum number of active satellites $\bar{R}$ that still guarantees both convergence and detection accuracy?*

B. Framework Architecture

To address the objectives in Section III-A, STELLAR integrates five core components (Fig. 1). The **Data Foundation Component** manages satellite traffic and states. The **Model Detection Component** implements a lightweight neural network for traffic recognition. To balance energy and convergence, the **Lightweight Adaptive Grouping Component** groups similar satellites to select representatives. The **Hierarchical Aggregation Component** executes a three-level update strategy (intra-orbital, ground station, global) with quality-aware weighting. Finally, the **Security Authentication Component** ensures participant trustworthiness via proxy-based verification.

Additionally, STELLAR employs an asynchronous aggregation protocol with staleness compensation to enhance robustness against intermittent LEO satellite links.

As shown in Fig. 1, the core learning modules form the federated learning processing unit. The Data Foundation Component supports the Model Detection and Adaptive Grouping Components with data; the latter two interact bidirectionally for coordinated training; the Adaptive Grouping Component passes results to the Hierarchical Aggregation Component for global model aggregation, which feeds back parameters to the Model Detection Component, forming a complete federated learning loop. The Security Authentication Component serves as an independent security assurance layer, authenticating nodes and encrypting communications throughout the framework.

The following subsections elaborate on the key components (excluding the data foundation).

C. Lightweight Detection Model

In STELLAR, representative nodes train the base model while others utilize it for detection. To balance recognition capability with resource constraints, we adopt a three-layer neural network architecture (Fig. 2) with only 3,682 parameters.

Parameterized as $f(x; w)$, the model processes 20-dimensional traffic features through an expand-then-contract structure ($20 \rightarrow 64 \rightarrow 32 \rightarrow 2$). It employs Batch Normalization (BN) for stability, ReLU activation for non-linearity, and Dropout to prevent overfitting. The network is trained by minimizing the standard cross-entropy loss. Future work may explore more advanced architectures, but this lightweight design currently meets the dual requirements of accuracy and efficiency.

D. Similarity-Based Representative Selection

Addressing Objectives 1 and 2. To reduce active satellites ($\bar{R}$) while maintaining convergence, we group satellites with similar data and select representatives. This simultaneously addresses:

- **Objective 1 (Energy)**: Reduces $\bar{R}$ from N to M (number of groups)
- **Objective 2 (Convergence)**: Bounded bias ensures convergence is preserved

Key Intuition. Satellites with similar traffic distributions tend to produce similar optimization signals. Therefore, one representative can approximate the update of a group of highly similar satellites, enabling us to reduce the number of active trainers while preserving convergence.

Formal Guarantee. This intuition is formalized in the theoretical analysis (Section III-I) via a similarity–gradient assumption and a representative-bias bound, which together justify similarity-based grouping and representative selection under Non-IID data and intermittent links.

1) Multi-Dimensional Similarity Metric: Direct distributional distance computation is infeasible. We infer via three proxies:

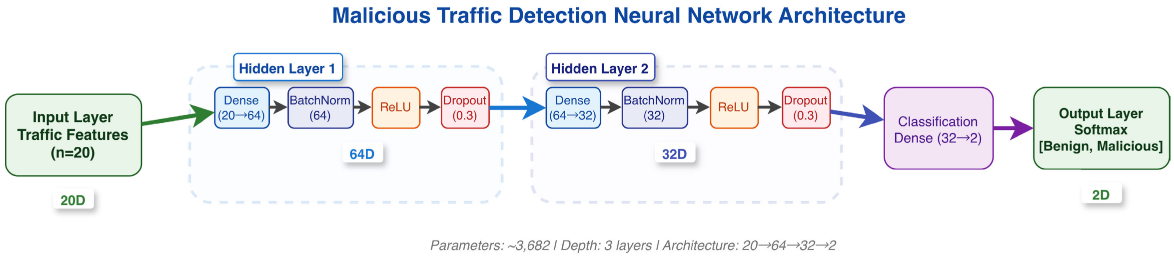


Fig. 2. Malicious traffic detection model architecture (20 → 64 → 32 → 2).

a) Model parameter similarity: Rationale: Similar data yields similar parameters.

Raw cosine similarity (not rescaled).

$$\mathrm{Sim}_{param}^{raw}(i,j) = \sum_{k=1}^{|\Theta|} w_k \cdot \frac{\mathrm{vec}(\theta_i^k)\cdot \mathrm{vec}(\theta_j^k)}{\|\mathrm{vec}(\theta_i^k)\|\cdot\|\mathrm{vec}(\theta_j^k)\|} \tag{3}$$

where θ_i^k and θ_j^k are k-th layer parameters of satellites i and j respectively, $|\Theta|$ is the number of model layers, $\mathrm{vec}(\cdot)$ denotes matrix vectorization operation, w_k is the weight coefficient for the k-th layer with $w_k \geq 0$ and $\sum_{k=1}^{|\Theta|} w_k = 1$, and deeper layers are typically assigned higher weights as they better reflect data distribution characteristics.

Rescaling to $[0,1]$ **(for consistency with other metrics).** Since the cosine similarity in Eq. (3) lies in $[-1,1]$, we linearly rescale it to $[0,1]$:

$$\mathrm{Sim}_{param}(i,j) = \frac{\mathrm{Sim}_{param}^{raw}(i,j)+1}{2} \tag{4}$$

For computational efficiency, model parameter similarity calculation is optimized by: (1) selecting key weight parameters while ignoring biases and batch normalization parameters; (2) normalizing parameters to reduce dimensional effects.

b) Training loss dynamics similarity: Rationale: Similar data distributions tend to induce similar *loss decrease trends*, which serve as a lightweight proxy for convergence behaviors.

Raw cosine similarity (not rescaled)

$$\mathrm{Sim}_{loss}^{raw}(i,j) = \frac{\sum_{e=1}^{E-1}(L_i^e - L_i^{e+1})(L_j^e - L_j^{e+1})}{\sqrt{\sum_{e=1}^{E-1}(L_i^e - L_i^{e+1})^2}\ \sqrt{\sum_{e=1}^{E-1}(L_j^e - L_j^{e+1})^2}} \tag{5}$$

where L_i^e is the local loss value of satellite i after the e-th local epoch and E is the number of local epochs.

Rescaling to $[0,1]$ **(for consistency with other metrics).** Since the cosine similarity in Eq. (5) lies in $[-1,1]$, we linearly rescale it to $[0,1]$:

$$\mathrm{Sim}_{loss}(i,j) = \frac{\mathrm{Sim}_{loss}^{raw}(i,j)+1}{2} \tag{6}$$

c) Prediction behavior similarity: Rationale: Similar data yields similar predictions.

Hellinger-based similarity (already in $[0,1]$**).**

$$\mathrm{Sim}_{pred}(i,j) = 1 - \frac{1}{K}\sum_{k=1}^{K}\frac{1}{\sqrt{2}}\cdot \sqrt{\sum_{c=1}^{C}\left(\sqrt{\hat{y}_{i,k,c}} - \sqrt{\hat{y}_{j,k,c}}\right)^2} \tag{7}$$

where $\hat{y}_{i,k,c}$ represents satellite i's model prediction probability for class c on the k-th test sample, K is shared test set size, C is the number of classes. This calculation, using Hellinger distance of prediction probability distributions, effectively captures differences in model prediction behavior.

d) Comprehensive similarity calculation: After normalizing all three similarity components to the range $[0,1]$, they are weighted and combined to yield comprehensive similarity:

$$\mathrm{Sim}(i,j) = \alpha\cdot \mathrm{Sim}_{param}(i,j) + \beta\cdot \mathrm{Sim}_{loss}(i,j) + \gamma\cdot \mathrm{Sim}_{pred}(i,j) \tag{8}$$

where α, β, and γ are weight coefficients satisfying $\alpha+\beta+\gamma = 1$. Based on experimental tuning, we set $\alpha = 0.6$, $\beta = 0.2$, $\gamma = 0.2$ to balance contributions of the three similarities.

This multi-dimensional metric integrates model parameters, loss dynamics, and prediction behaviors to capture data distribution similarity. The normalization ensures $\mathrm{Sim}(i,j) \in [0,1]$, consistent with Assumption A3 in Section III-I.

2) Adaptive Satellite Grouping: **Design Rationale.** STELLAR adopts lightweight greedy grouping rather than complex AI schedulers (e.g., DRL). While heavy schedulers might offer theoretical optimality, their on-board inference energy would negate FL savings. Our data-driven threshold adaptation (Eq. 9) achieves adaptivity with minimal overhead.

Based on the multi-dimensional similarity, we propose a systematic grouping strategy. Initially, a coordinator node with optimal ground station visibility is selected for each orbit. The system initializes with each satellite as an independent group, setting a maximum neighborhood search distance (d_{max}), a group size threshold (G_t), and an initial similarity threshold ($\theta_i = 0.8$).

The grouping process employs a greedy strategy: the coordinator sequentially processes unvisited satellites as "source nodes" i, searching for candidate neighbors j within $|i-j| \leq d_{max}$. If an ungrouped node j satisfies $\mathrm{Sim}(i,j) \geq \theta_i$, it is absorbed into group i. Even if j belongs to another group, it is reallocated to group i if it exhibits higher similarity. The search terminates when consecutive nodes cannot join. This strategy recalculates similarity and updates groupings every T_{update} rounds to adapt to topology changes.

To address the limitation of static scheduling, we introduce a **Data-Driven Threshold Adaptation** mechanism. The similarity threshold θ is dynamically updated based on real-time group statistics to prevent groups from becoming too loose or strict as data distributions drift:

$$\theta_{new}(i) = \max\big(0.6, \min(1.0,$$

$$\min_{k \in G_i} \mathrm{Sim}(i,k))\Big) \quad \text{if } |G_i| \geq G_t. \tag{9}$$

Additionally, we enforce a **Participation Control** constraint, $|S_{active}| \leq S_{max}$, to prevent system overload in high-density orbits.

3) Representative Node Selection: Upon successful group formation, the source node i serves as the initial representative node, responsible for the group's first model training and parameter communication tasks. To achieve balanced resource utilization and extend satellite operational lifespans, representative nodes are periodically rotated to prevent energy depletion from continuous training loads.

Representative node rotation occurs in non-grouping refresh rounds as follows:

1) **Rotation sequence establishment**: Group members form an ordered rotation sequence based on their positions, with the source node as the initial representative.
2) **Clockwise rotation mechanism**: The representative role transfers sequentially to the next member in the predefined order, cycling back to the first member after the last.
3) **Energy-aware selection**: Before assignment, the system checks that the candidate node has sufficient energy for training and communication. If energy is insufficient, the next node is evaluated. If no energy-sufficient node is found after checking all group members, the group will have no active representatives and thus not participate in training or model aggregation in this round.

This rotation mechanism ensures load balancing and system stability via energy-aware representative selection, preventing premature node failures from resource exhaustion.

4) Intra-Group Model Training: After group formation, the representative node conducts model training while others directly utilize the result, significantly reducing computational overhead.

The **representative node** executes the complete training lifecycle: (1) receiving current global model parameters from the coordinator; (2) performing local training via gradient descent on the local dataset; (3) generating model updates for aggregation; and (4) distributing the updated parameters to group members.

In contrast, **non-representative members** operate in a passive mode, primarily responsible for: (1) synchronizing model parameters from the representative; (2) executing inference tasks directly; and (3) periodically broadcasting energy status to facilitate rotation decisions. This distinct division of labor maximizes computational efficiency and prolongs the operational lifespan of individual satellites.

E. Hierarchical Aggregation Mechanism

Addressing Objective 2 (Convergence under Non-IID Data). To maintain convergence despite data heterogeneity, we design quality-aware weighting that balances group size, data volume, and local accuracy. This prevents low-quality updates from dominating the global model, ensuring convergence in Non-IID scenarios.

To reduce communication overhead while ensuring aggregation effectiveness, a three-level hierarchical aggregation mechanism is designed: intra-orbital aggregation, ground station aggregation, and global aggregation.

1) Intra-Orbital Aggregation: Orbital coordinators collect model updates from representative nodes of all groups within the same orbit and perform weighted average aggregation. The aggregation weights consider three factors: (1) Group size—larger groups receive higher weights reflecting more represented satellites; (2) Training data volume—groups with more training samples receive higher weights; (3) Model performance—groups with higher local validation accuracy receive higher weights.

The intra-orbital aggregation is formulated as:

$$\Delta w_{orbit} = \sum_{i=1}^{M} \alpha_i \cdot \Delta w_i \tag{10}$$

where Δw_{orbit} is the aggregated orbital update, Δw_i is the i-th group's model update, M is the number of groups in the orbit, and α_i is the normalized weight:

$$\alpha_i = \frac{\tilde{G}_i \cdot \tilde{D}_i \cdot \widehat{\mathrm{ACC}}_i}{\sum_{j=1}^{M} \tilde{G}_j \cdot \tilde{D}_j \cdot \widehat{\mathrm{ACC}}_j} \tag{11}$$

It is worth noting that while our theoretical analysis assumes static weights (proportional to data volume) for mathematical tractability, the practical implementation employs the dynamic quality-aware weights defined in Eq. (11). Empirically, this strategy acts as a variance reduction technique by down-weighting low-quality updates from unstable satellites, further enhancing convergence stability beyond the theoretical lower bounds.

2) Ground Station and Global Aggregation: Ground stations aggregate orbital updates from their assigned orbits (typically 1-2 orbits per station):

$$\Delta w_{station} = \sum_{i=1}^{m} \beta_i \cdot \Delta w_{orbit,i} \tag{12}$$

where m is the number of orbits per station, and β_i weights are proportional to orbital data volumes.

Finally, global aggregation combines all station updates:

$$\Delta w_{global} = \sum_{i=1}^{s} \gamma_i \cdot \Delta w_{station,i} \tag{13}$$

where s is the number of ground stations, and γ_i weights reflect station data contributions.

This hierarchical architecture leverages the satellite network's natural topology, minimizing inter-satellite communication while maintaining aggregation quality through contribution-based weighting at each level.

F. Asynchronous Aggregation and Staleness Compensation

Enhancing Robustness against Link Instability. Recognizing the inherent intermittency and high latency of LEO satellite links, STELLAR incorporates a time-window-based asynchronous aggregation protocol equipped with staleness compensation. This mechanism serves as a critical resilience layer, operating synergistically with the quality-aware aggregation (Section III-E) to ensure continuous training progress and system survivability, even when communication links experience severe delays or interruptions.

1) Time-Window Asynchronous Protocol: Unlike synchronous FL protocols that must wait for all stragglers, STELLAR imposes a strict maximum waiting window ΔT for each aggregation round. Updates arriving within this window are aggregated immediately, while late updates are buffered and treated as "stale" in subsequent rounds.

- **Intra-orbital Window**: $\Delta T_{\text{orbit}} = 30\,\text{s}$, accounting for local training latency and ISL transmission delays.
- **Ground Station Window**: $\Delta T_{\text{gs}} = 60\,\text{s}$, adapted to the limited visibility duration of LEO satellites relative to ground stations.

Updates that miss the current window are preserved for the next available round, tagged with a specific staleness value τ.

2) Staleness Compensation: Stale updates may carry gradients derived from obsolete model parameters, potentially introducing noise. To leverage this data while mitigating adverse effects, we define the staleness of an update from node i as $\tau_i = t_{\text{curr}} - t_{\text{gen}}$, where t_{curr} is the current aggregation time and t_{gen} is the timestamp when the node's local model was generated. In our theoretical analysis, staleness is measured in *rounds*; the time-window delays can be mapped to a bounded round-level delay $\tau \leq \tau_{\max}$ (Assumption A4). We apply an **Inverse Time Decay** function to weight these updates:

$$\beta(\tau) = \frac{1}{1 + \lambda \cdot \tau} \tag{14}$$

where λ is a tunable decay coefficient. This mechanism ensures that fresh updates ($\tau \to 0, \beta \approx 1$) contribute fully, whereas stale updates are down-weighted rather than discarded, thereby maximizing data utilization under unstable conditions. Theoretical analysis (Section III-I) further demonstrates that the error introduced by this asynchronous mechanism is bounded and controllable.

G. Lightweight Satellite Authentication Based on Proxy Nodes

Addressing Security Requirements. The security of satellite identities and parameter passing is crucial for federated learning. Given resource-constrained space environments and intermittent connectivity, we design LASS-PN (Lightweight Authentication for Satellites based on Proxy Nodes) to ensure participant trustworthiness with minimal overhead.

LASS-PN verifies satellite legitimacy, generates communication keys, and enables authentication among all satellites via cryptographic techniques and proxy node mechanisms. LASS-PN selects Proxy Authentication Satellites (PAS) for professional authentication handling. Normal Satellites (NS) only interact with PAS to authenticate trust, freeing them from cumbersome processes and significantly reducing overhead.

LASS-PN's core cryptographic modules use coordinated SM3 and SM4 algorithms: Authentication uses an SM3-256 hash engine, with *MAC* generation function $f1$ truncating the first 128 bits of hashes to generate *MAC*, and response function $f2$ extracting the last 128 bits as RES. Key derivation function *KDF* employs SM3-HMAC-256, truncating the first 128 bits of 256-bit output as encryption keys. Data encryption *enc* uses 128-bit key SM4, ensuring confidentiality via fixed-length ciphertext with security equivalent to AES-128.

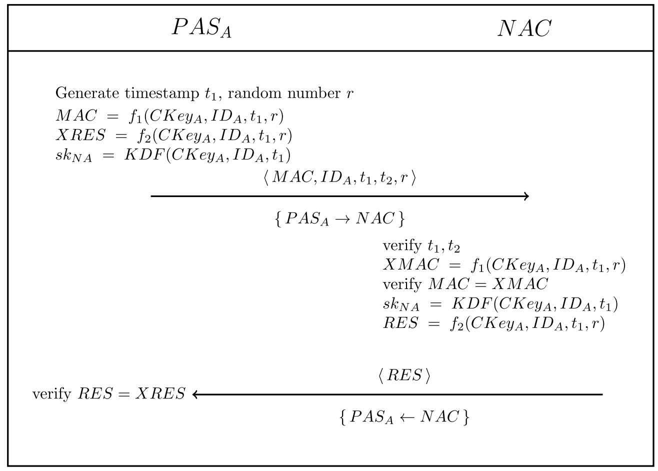


Fig. 3. Proxy node satellite-ground authentication process.

1) Node Registration: Network Authentication Center (NAC) assigns parameters to satellites (PAS and NS) via out-of-band secure channels. The parameters include identity identifiers *ID*, secrets *CKey*, intra-group pre-shared keys *GKey*, and NS identity keys *IDKey*.

For PAS registration, NAC generates identifier ID_A for satellite A, designates A as PAS according to the grouping strategy, gets current timestamp t_s, and generates PAS identity secret $CKey_A$ using NAC private key s as follows:

$$CKey_A = KDF(s, ID_A, t_s). \tag{15}$$

NAC generates $GKey_A$ as intra-group identity shared key of PAS_A and intra-group normal nodes, which is used by PAS to authenticate the group identity of intra-group NS. NAC sends $CKey_A$ and $GKey_A$ to PAS through secure channels.

For NS registration, NAC assigns unique identifier ID_i to NS_i, assigns NS_i to the group of nearby proxy PAS_A based on satellite orbital parameters and topological visibility. NAC generates NS_i's identity secret $CKey_i$ using its private key s, and sends it along with the group's intra-group identity shared key $GKey_A$ and group member IDs to NS_i through secure channels.

2) Proxy Node Authentication: First, PAS and NAC perform satellite-ground authentication and negotiate session key sk_{NA}. Then, Adjacent PAS authentication is conducted.

a) Satellite-ground authentication: NAC authenticates PAS legitimacy; upon completion, it generates PAS-NAC communication key sk_{NA}. Steps are in Fig. 3.

Step 1: PAS_A gets current timestamp t_1, generates random number r, and produces MAC, authentication response $XRES$, and $PAS_A - NAC$ communication key sk_{NA} via formulas as in Fig. 3.
After generating these parameters, PAS_A adds current timestamp t_2 and sends authentication request containing the content $\langle MAC, ID_A, t_1, t_2, r \rangle$ to NAC.

Step 2: NAC parses the request, extracts parameters, checks ID_A legitimacy and timestamp validity ($t_0 - t_2 \leq \Delta t$), with t_0 as current time and Δt maximum transmission time). If valid, it queries satellite info via ID_A; if none

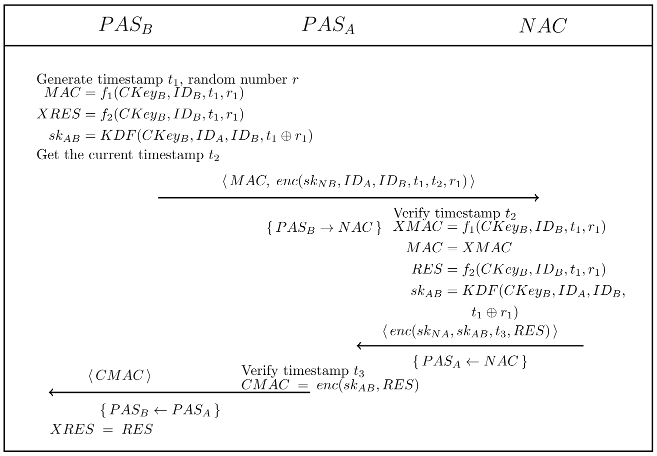


Fig. 4. Adjacent proxy node authentication process.

exists, it deems the satellite illegal and stops verification. Otherwise, it extracts $CKey_A$, gets r and t_1 from the message, and calculates message verification code $XMAC$.

If verification passes, PAS_A is authenticated. NAC generates sk_{NA} and response code RES via formulas as in Fig. 3, then sends RES to PAS_A.

Step 3: PAS_A verifies response code consistency with $XRES$; if valid, it completes identity authentication and session key negotiation with NAC. PAS_A and NAC then use sk_{NA} for secure communication.

Step 4: After PAS authentication completion, NAC sends network-wide satellite identifiers and grouping correspondences $\langle IDs, PAS\rangle$ to each PAS via the negotiated session key sk_{NA}. Each PAS stores locally for subsequent intra-group satellite authentication and rapid determination of normal node groups and corresponding PAS during inter-group authentication communication.

Simultaneously, NAC generates NS_i identity key $IDKey_i$ using NS_i identity secret $CKey_i$ and corresponding PAS_A identity identifier ID_A:

$$IDKey_i = KDF(CKey_i, ID_A). \tag{16}$$

NAC sends $\langle ID, IDKey\rangle$ to corresponding PAS_A through negotiated session key sk_{NA}. PAS utilizes $IDKey$ to verify NS identity during intra-group authentication.

b) Adjacent proxy node authentication: Adjacent proxy nodes need to interact and assist in completing authentication between inter-group members. Therefore, proxy nodes first authenticate each other's identities. This authentication is completed collaboratively by relevant proxy nodes and NAC. The process is shown in Fig. 4.

Step 1: Assume authentication is initiated by proxy satellite PAS_B. PAS_B first generates random number r_1, timestamp t_1, and generates MAC, expected response value $XRES$, and inter-proxy communication key sk_{AB} as in Fig. 4. PAS_B obtains current timestamp t_2 and sends authentication request $\langle MAC, enc(sk_{NB}, ID_A, ID_B, t_1, t_2, r_1)\rangle$ to NAC.

Step 2: NAC parses received information, verifies validity of ID_B and ID_A, and determines whether timestamp t_2 satisfies $t_0 - t_2 \le \Delta t$. Next, NAC queries corresponding identity information based on identifier ID_B. If satellite is unregistered, authentication terminates; otherwise, $XMAC$ is generated using $CKey_B$ as in Fig. 4 to verify MAC legitimacy.

NAC then calculates response value RES and inter-proxy communication key sk_{AB} as in Fig. 4.

NAC obtains current timestamp t_3, encrypts t_3, RES, sk_{AB} using sk_{NA} and sends to PAS_A as in Fig. 4.

Step 3: PAS_A decrypts the message using sk_{NA} and verifies timestamp $t_0 - t_3 \le \Delta t$. After verification passes, it encrypts RES using sk_{AB} to get $CMAC$ and sends to PAS_B as in Fig. 4.

Step 4: PAS_B decrypts $CMAC$ using sk_{AB} to obtain parameter RES, verifies consistency between $XRES$ and RES. If verification passes, PAS_B completes authentication of PAS_A.

3) Intra-Group Node Authentication: Intra-group authentication must be completed before normal satellite nodes communicate. The process is as shown in Fig. 5.

Step 1: Normal satellite NS_i generates key $IDKey_i$ using registration phase identity secret $CKey_i$ and its PAS node PAS_A identifier ID_A via formula 16, then encrypts timestamp t_1 with it, and sends the authentication request $\langle ID_i, enc(IDKey_i, t_1, ID_i)\rangle$ to PAS_A.

Step 2: PAS_A receives authentication request from ID_i, it determines whether ID_i is its group member assigned by NAC. After confirming that, it extracts $IDKey_i$ through $\langle ID, IDKey\rangle$ correspondence obtained during PAS authentication phase to decrypt and verify ID_i. Finally, it verifies timestamp $t_0 - t_1 \le \Delta t$.

After verification passes, PAS_A generates random number r_1, timestamp t_2, and generates MAC along with $GKey_A$. Then it generates encrypted timestamp protection sequence AK and intra-group session key sk_A. PAS_A encrypts authentication message $\langle ID_A, MAC, enc(IDKey_i, r_1, t_1, t_2 \oplus AK)\rangle$ with $IDKey_i$ to send to node NS_i.

Step 3: NS_i receives returned message from PAS_A, decrypts with $IDKey_i$, generates $XMAC$ to verify legitimacy.

After verification passes, NS_i parses message parameters and generate intra-group communication key sk_A.

This process repeats for other group members, ultimately establishing secure intra-group communication via shared key sk_A.

4) Inter-Group Node Authentication: Edge satellites of adjacent groups cannot complete mutual authentication via intra-group authentication; the authentication is assisted by the corresponding group proxy satellites.

Given normal satellites NS_i (of PAS_A group) and NS_j (of PAS_B group), NS_i seeks to authenticate NS_j, the process is as in Fig. 6.

Step 1: NS_i gets the timestamp t_1 and sends to PAS_A an ID_j-included authentication request encrypted via intra-group key sk_A.

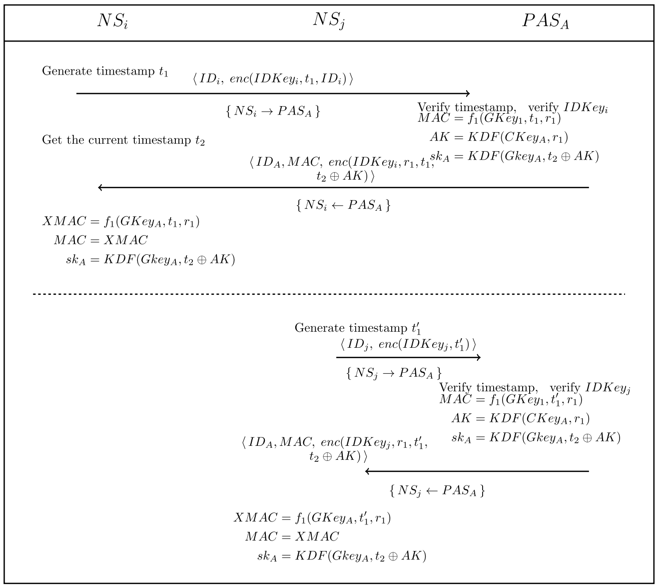


Fig. 5. Intra-group satellite authentication process.

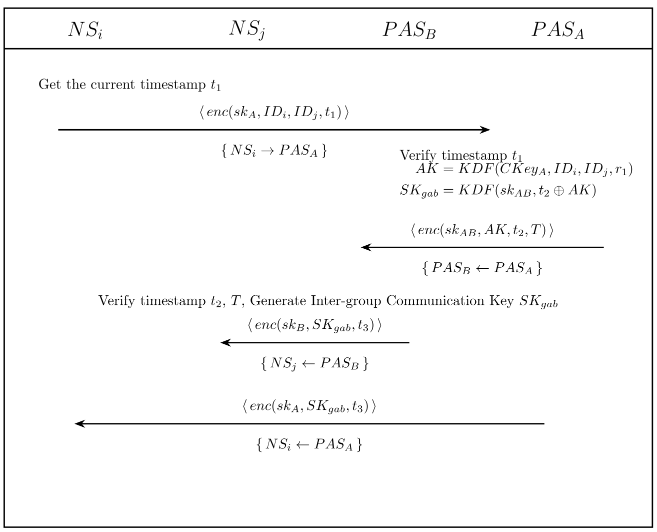


Fig. 6. Inter-group satellite authentication process.

Step 2: PAS_A decrypts the request, verifies parameters, finds PAS_B via ID_j, generates r, then creates SK_{gab} using $CKey_A$, r, ID_i, and ID_j, sets the expiration time T. ID_i and ID_j are used to generate the inter-group session key to enhance its randomness, not to limit it to ID_i-ID_j communication. PAS_A encrypts and sends AK, $t2$, T to PAS_B via inter-proxy key sk_{AB}, generates timestamp t_3, and sends inter-group key SK_{gab} to intra-group nodes using intra-group key sk_A.

Step 3: PAS_B verifies $t2$, T, etc. Upon success, it generates inter-group key SK_{gab}, and sends it to intra-group nodes via sk_B.

H. STELLAR Deployment

Fig. 7 shows STELLAR's deployment. All nodes authenticate via LASS-PN before communication, establishing trust and secure keys.

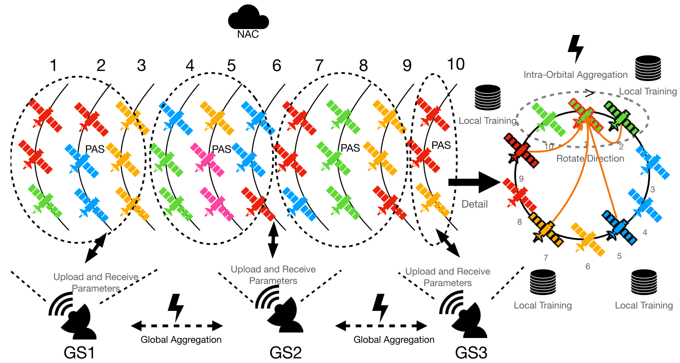


Fig. 7. Stellar deployment diagram.

The upper right details a 10-satellite orbit: satellites form **authentication groups** (dashed boxes) by visibility, managed by fixed PAS nodes, and **computational groups** (colored) by data similarity for federated learning. Each computational group selects a representative (bold) for training; others adopt its model. Representatives rotate (dashed arrows) for load balancing. During aggregation, representatives send updates (orange lines) to orbital coordinators, implementing three-level aggregation: intra-orbital → ground station → global.

I. Theoretical Analysis

This section provides rigorous guarantees for STELLAR's design objectives. To strictly analyze the impact of our proposed *Similarity-based Representative Selection* and *Asynchronous Staleness Compensation*, we formulate the analysis under the Asynchronous Stochastic Gradient Descent (Async-SGD) setting. While our implementation uses Local SGD (FedAvg) for communication efficiency, standard FL theory confirms that local updates introduce only an additional bounded drift term [23]. Our proofs focus on bounding the unique error terms introduced by STELLAR: the *representative bias* and the *asynchronous staleness*.

1) Problem Setting: Consider N satellites $\mathcal{V} = \{1, 2, \ldots, N\}$, each holding local data $D_v = \{(x_i^{(v)}, y_i^{(v)})\}_{i=1}^{n_v}$. The local objective is

$$F_v(w) = \frac{1}{n_v}\sum_{i=1}^{n_v} \ell(w; x_i^{(v)}, y_i^{(v)}), \quad F(w) = \sum_{v=1}^{N} p_v F_v(w), \tag{17}$$

where $p_v = n_v / \sum_u n_u$ and $\ell(\cdot)$ is the local loss. STELLAR activates only a subset of satellites $\mathcal{R}_t$ (representatives) at each round t.

The total per-round energy is

$$E_t^{\text{tot}} = |\mathcal{R}_t| e_{\text{act}} + (N - |\mathcal{R}_t|) e_{\text{pas}}, \tag{18}$$

where e_{act} and $e_{\text{pas}} = \varepsilon e_{\text{act}}$ are active and passive energy costs. Thus, total energy over T rounds is $E_{\text{total}} \approx T\bar{R} e_{\text{act}}$, implying that minimizing the number of active representatives $\bar{R}$ directly improves energy efficiency.

2) Assumptions:

- **A1 (Smoothness).** Each F_v is L–smooth: $\|\nabla F_v(w) - \nabla F_v(w')\| \le L\|w - w'\|$.
- **A2 (Bounded Variance).** $\mathbb{E}\|\nabla \ell(w; \xi) - \nabla F_v(w)\|^2 \le \sigma^2$.

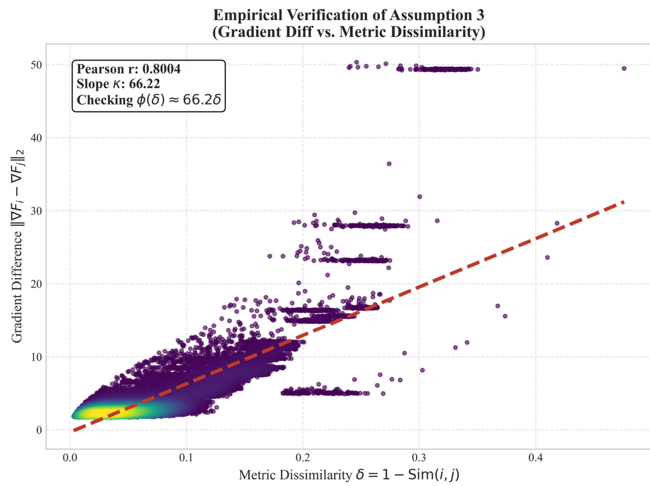


Fig. 8. Empirical validation of A3. Strong correlation ($r = 0.80$) between metric dissimilarity δ and gradient distance supports the proposed multi-dimensional similarity metric.

- **A3 (Similarity–Gradient Relationship).** There exists a non-decreasing function $\phi : [0, 1] \to \mathbb{R}_+$ with $\phi(0) = 0$ such that for any satellites i, j,

$$\|\nabla F_i(w) - \nabla F_j(w)\| \le \phi(1 - \mathrm{Sim}(i, j)). \tag{19}$$

Empirical Evidence. At round 6 (after similarity-based grouping is activated), we compute $\delta_{ij} = 1 - \mathrm{Sim}(i, j)$ and the corresponding gradient distance for all 208,335 satellite pairs in the simulated constellation, and observe a strong positive correlation (Pearson $r = 0.80$, $p < 0.001$), indicating that larger metric dissimilarity aligns with larger gradient discrepancy (Fig. 8). Using linear regression, we fit a conservative linear form $\phi(\delta) \approx 66.2\delta$. At the design threshold $\theta = 0.8$ (i.e., $\delta_{\max} = 0.2$), this yields a worst-case bound $\phi(0.2) \approx 13.2$. In practice, the *intra-group pairs produced by our grouping procedure* typically exhibit much smaller dissimilarity (often $\delta \ll 0.2$), leading to substantially tighter observed gradient gaps and providing a safety margin in our deployment setting.
- **A4 (Bounded Staleness).** Delays $\tau_{t,r}$ satisfy $0 \le \tau_{t,r} \le \tau_{\max}$.
- **A5 (Bounded Gradient).** The stochastic gradients are almost surely bounded: $\|\nabla \ell(w; \xi)\| \le G$, $\forall w, \xi$.

3) Representative Approximation: Satellites are partitioned into similarity groups $\{\mathcal{G}_m\}_{m=1}^{M_t}$, each represented by $r_m \in \mathcal{G}_m$. Let $P_m = \sum_{j \in \mathcal{G}_m} p_j$ denote the aggregate weight of group m. Since grouping may change over rounds, define the global minimum intra-group similarity as $\theta \triangleq \min_t \min_m \theta_m^{(t)}$, where $\theta_m^{(t)} = \min_{i,j \in \mathcal{G}_m^{(t)}} \mathrm{Sim}(i, j)$.

We define two key quantities:

- *Group-average gradient*: $g_m(w) = \sum_{j \in \mathcal{G}_m} \frac{p_j}{P_m} \nabla F_j(w)$, which is the true weighted gradient of all satellites in group m.
- *Representative gradient*: $g_{\mathrm{rep}}(w) = \sum_{m=1}^{M_t} P_m \nabla F_{r_m}(w)$, which aggregates only gradients from representatives.

The core question is: how well does $g_{\mathrm{rep}}(w)$ approximate the true global gradient $\nabla F(w)$?

Lemma 1: [Representative Bias Control]

For each group $\mathcal{G}_m$ with representative r_m and minimum intra-group similarity $\theta_m = \min_{i,j \in \mathcal{G}_m} \mathrm{Sim}(i, j)$,

$$\|g_m(w) - \nabla F_{r_m}(w)\| \le \phi(1 - \theta_m).$$

Consequently, letting $\theta = \min_t \min_m \theta_m^{(t)}$,

$$\|g_{\mathrm{rep}}(w) - \nabla F(w)\| \le \phi(1 - \theta).$$

Remark: This lemma formalizes the intuition in Section III-D that similarity-based grouping introduces bounded bias, validating the representative mechanism for energy reduction while preserving convergence.

4) Asynchronous Aggregation: The global model is updated as $w_{t+1} = w_t - \eta \hat{g}_t$, where $\eta > 0$ is the learning rate.

STELLAR adopts a time–window protocol where late updates are weighted by

$$\beta(\tau) = \frac{1}{1 + \lambda\tau}, \quad \lambda \in \big(0, 1\big]. \tag{20}$$

We use $g_{r_m}(w)$ to denote a stochastic gradient computed by representative r_m at model parameter w, with $\mathbb{E}[g_{r_m}(w)] = \nabla F_{r_m}(w)$.

We denote by $\hat{g}_t$ the staleness-compensated aggregated gradient at round t:

$$\hat{g}_t = \sum_{m=1}^{M_t} P_m \beta(\tau_{t,m}) \, g_{r_m}(w_{t-\tau_{t,m}}),$$

Lemma 2: [Asynchronous Error Bound]

Under Assumptions A1–A5,

$$\mathbb{E}\|\hat{g}_t - g_{\mathrm{rep}}(w_t)\|^2 \le 4G^2 \frac{\tau_{\max}^2}{(1 + \lambda\tau_{\max})^2} + 4L^2\eta^2 G^2 \tau_{\max}^2 + 2\sigma^2. \tag{21}$$

Thus, the asynchronous effect is bounded and decreases as λ increases (for $\lambda \in (0, 1]$).

5) Main Results:

Theorem 1: [Convergence Guarantee (Objective 2)]

Under A1–A5 and $\eta = \mathcal{O}(1/\sqrt{T})$,

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\|\nabla F(w_t)\|^2 \le \mathcal{O}\Big(\tfrac{1}{\sqrt{T}}\Big) + \mathcal{O}(\sigma^2) + \mathcal{O}\big(\phi(1-\theta)^2\big) + \mathcal{O}\big(\Psi(\tau_{\max}, \lambda)\big). \tag{22}$$

where $\Psi(\tau_{\max}, \lambda) = \mathcal{O}\left(\frac{\tau_{\max}^2}{(1+\lambda\tau_{\max})^2}\right)$ is the asynchronous error term from Lemma 2. Hence, STELLAR achieves the same $\mathcal{O}(1/\sqrt{T})$ convergence rate as full-participation FL, with controlled bias. Moreover, it converges to a neighborhood of stationary points whose radius is determined by σ^2, $\phi(1-\theta)^2$, and $\Psi(\tau_{\max}, \lambda)$.

Theorem 2: [Energy Saving Guarantee (Objective 1)]

Compared with FedAvg, STELLAR reduces energy consumption by

$$\rho_{\mathrm{save}} = (1 - \varepsilon)\left(1 - \frac{\bar{R}}{N}\right), \tag{23}$$

where $\varepsilon = e_{\mathrm{pas}}/e_{\mathrm{act}} \ll 1$. In our simulated configuration ($N = 646$, $\bar{R} \approx 150$), this predicts $\rho_{\mathrm{save}} \approx 73\%$, consistent with the measured 77–80% reduction.

Proof Sketch: The analysis proceeds as follows. (1) By Lemma 1, the representative gradient approximates the true gradient with bias $\phi(1-\theta)$. (2) Lemma 2 bounds the staleness error using $\beta(\tau)$ decay. (3) Substituting both bounds into the smoothness-based descent inequality and telescoping over T iterations yields the final rate. (4) Energy efficiency follows from the reduced number of active satellites per round.

Remark: Detailed proofs and intermediate derivations are provided in the supplementary material.

IV. Performance Evaluation

A. Experimental Environment and Dataset

The experimental network simulates a large-scale LEO satellite network based on the **OneWeb constellation**. The orbital parameters are initialized using public TLE data. Although the raw TLE set contains 651 satellite entries, we filter out inactive/standby units, resulting in **646 active satellites** distributed across 12 orbital planes. The constellation operates at an altitude of approximately 1200 km with an inclination of 87.9°.

The simulation core uses the Skyfield library to calculate real-time satellite positions and velocities via TLE data. **Simulation Fidelity:** Our environment models the *dynamic topology evolution* of the OneWeb constellation. Satellite positions and inter-satellite link (ISL) visibility are updated every second based on orbital mechanics. **Unless otherwise stated, the main experiments assume ideal communication links (no stochastic delay or packet loss) to isolate algorithmic gains.** We acknowledge this represents a best-case scenario; Section IV-E partially addresses this limitation by validating system robustness under severe delays (up to 60% link instability).

Each satellite has 1000 Wh batteries and 2.5 m^2 solar panels (30% conversion efficiency). Energy consumption is dynamically calculated by actual tasks: transmission energy depends on data size and bandwidth; computational energy is proportional to training operations.

This paper utilizes satellite network traffic data from the STIN-IDS dataset [12]. To adapt the dataset for binary intrusion detection, we merge the original labels: the 'Benign' class is mapped to *Normal*, and all 7 attack types are unified as *Malicious*. After preprocessing, the dataset contains **2,122,322 records**, partitioned into training and testing sets with an 8:2 ratio.

To rigorously evaluate the proposed framework, we design two data distribution scenarios:

1) **Independent and Identically Distributed (IID)**: Data is randomly shuffled and uniformly distributed across all satellites, ensuring that each node holds samples with similar feature distributions.
2) **Non-Independent and Identically Distributed (Non-IID)**: Unlike the standard Dirichlet partition, we construct a *feature-skewed* non-IID scenario based on physical geographic mappings to reflect realistic satellite coverage:
   - *Inter-orbit Feature Shift*: Satellites in different orbits cover distinct geographic regions (e.g., ocean vs. urban), resulting in significant differences in traffic statistical features (Feature Skew).
   - *Intra-orbit Data Overlap*: Considering the temporal correlation of satellites in the same orbital plane, we introduce a data overlap mechanism (overlap ratio set to 0.5) to simulate the shared traffic characteristics of adjacent satellites during orbital movement.

The data allocation strictly follows the physical topology of the OneWeb constellation to validate STELLAR's robustness against realistic heterogeneity.

B. Evaluation Metrics

To comprehensively assess the performance of the proposed framework, we utilize the following metrics:

- **Accuracy**: Proportion of correctly classified samples;
- **Precision**: Ratio of true positives to total positive predictions;
- **Recall**: Ratio of true positives to actual positive samples;
- **F1-Score**: Harmonic mean of precision and recall;
- **Energy Consumption**: Total energy consumption per training round;
- **Participating Satellites**: Number of satellites actively participating in training per round;
- **Energy Efficiency**: Ratio of accuracy to energy consumption;
- **Training Efficiency**: Convergence speed measured by rounds to reach target accuracy.

C. Comparison Methods

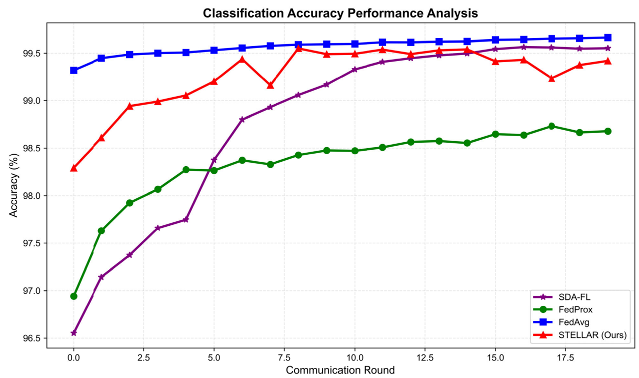
STELLAR is compared with the following baselines:

1) **FedAvg**: The standard Federated Averaging algorithm, where participating satellites upload locally trained model parameters, and the server aggregates them via weighted averaging.
2) **FedProx**: An improved framework based on FedAvg that introduces a **proximal term** to the local objective function. It limits the deviation between local and global models, theoretically designed to address system heterogeneity and Non-IID data challenges.
3) **SDA-FL**: A **specialized** federated learning framework tailored for satellite networks, which integrates **semi-supervised domain adaptation**. It utilizes GANs to generate synthetic data and incorporates iterative pseudo-labeling mechanisms to align feature distributions across heterogeneous satellites.

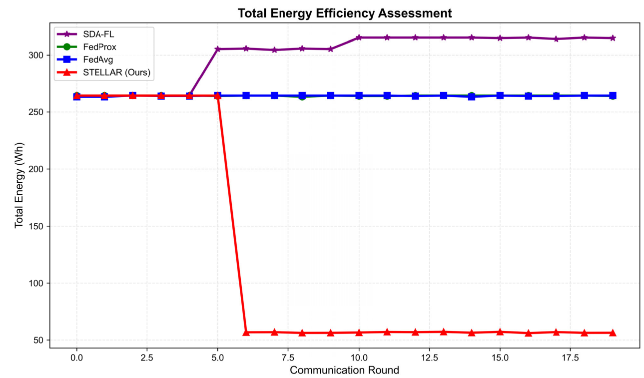
All methods are implemented in the same OneWeb simulation environment with identical hyperparameters to ensure a fair comparison.

D. Experimental Results and Analysis

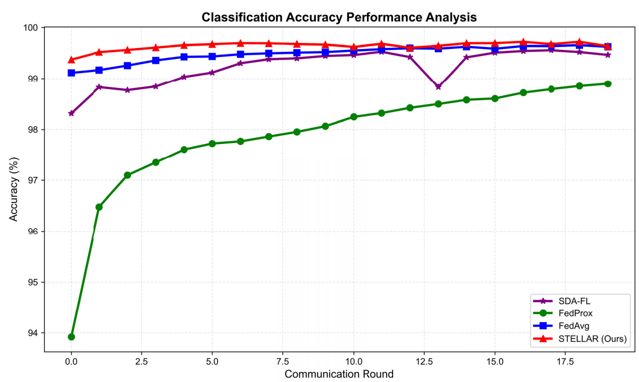
1) Performance in IID and Non-IID Environments: Fig. 9 presents the comprehensive performance comparison across both data distribution scenarios.



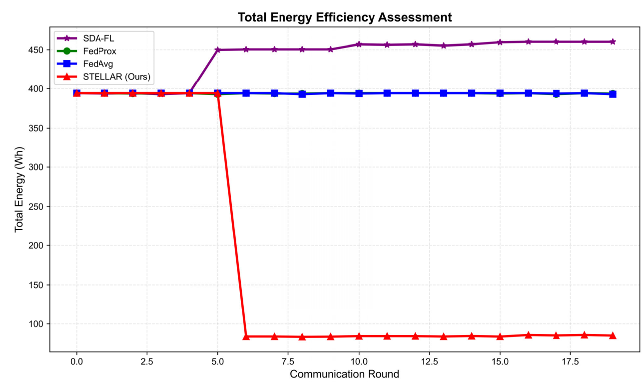
(a) IID: Accuracy



(b) IID: Energy Consumption



(c) Non-IID: Accuracy



(d) Non-IID: Energy Consumption

Fig. 9. Accuracy and energy consumption comparison in IID and Non-IID environments. Top row: IID scenario where STELLAR achieves 99.55% accuracy with 77% energy savings. Bottom row: Non-IID scenario where STELLAR reaches 99.72% accuracy with 80% energy savings. The sharp energy drop at Round 6 corresponds to the activation of similarity-based grouping, reducing active satellites from 646 to approximately 150-170.

a) IID environment: From Fig. 9a, STELLAR achieves detection accuracy comparable to the best baseline FedAvg, maintaining over 99.5% in the later stages. SDA-FL starts with lower accuracy but gradually catches up, while FedProx shows relatively weaker performance (~98.7%) and slower convergence. Notably, despite actively using only a fraction of the constellation after Round 6, STELLAR maintains this competitive accuracy, demonstrating that the representative node mechanism effectively captures the global data distribution without requiring full participation.

The resource efficiency advantage of STELLAR is drastic, as shown in Fig. 9b. Before Round 6 (warm-up phase), all methods consume similar energy (~260 Wh). However, once the similarity-based grouping is activated at Round 6, STELLAR's energy consumption **plummets to approx. 60 Wh**, resulting in a **77% reduction** compared to baselines. This energy reduction corresponds to a sharp drop in active satellites from 646 to roughly 150, achieving an energy efficiency (Accuracy/Energy) improvement of 4× (1.7 vs. 0.4%/Wh). In contrast, SDA-FL consumes the highest energy (~310 Wh) due to the computational overhead of GAN training. In terms of communication overhead, STELLAR reduces per-round model uploads from 646 to approximately 150, corresponding to a 77% reduction in transmitted data volume.

b) Non-IID environment: In the more challenging Non-IID scenario (Fig. 9c), STELLAR demonstrates superior robustness, with its accuracy curve consistently outperforming other methods and stabilizing near 99.8%. FedAvg follows closely, while SDA-FL exhibits instability and high variance. Most notably, **FedProx performs the worst**, contradicting theoretical expectations. This is attributed to the *feature skew* nature of satellite data: the proximal term in FedProx restricts local models from adapting to the distinct feature distributions of different orbital regions (e.g., ocean vs. urban), leading to over-regularization. STELLAR's grouping strategy avoids this by allowing similar nodes to collaborate freely.

The energy advantage of STELLAR becomes even more critical in Non-IID settings, as shown in Fig. 9d. Baseline methods consume significantly more energy (~400-450 Wh) compared to the IID scenario due to the difficulty of convergence. However, STELLAR maintains its low-energy profile, dropping to **~80 Wh** after Round 6. This represents an **80% energy saving** compared to SDA-FL and FedProx. With active satellites reduced from 646 to approximately 170, STELLAR achieves an energy efficiency of ~1.2%/Wh, whereas other methods struggle at ~0.25%/Wh—a 5× **improvement**. This confirms that STELLAR provides a sustainable solution for massive satellite constellations where energy is a scarce resource.

2) Performance Comparison Under Different Data Distributions: To comprehensively evaluate method adaptability under different data distribution conditions, we compare final

TABLE I
PERFORMANCE COMPARISON OF METHODS UNDER IID AND NON-IID DISTRIBUTIONS

Method	Accuracy (%)		F1-Score (%)		Precision (%)		Recall (%)	
	IID	Non-IID	IID	Non-IID	IID	Non-IID	IID	Non-IID
SDA-FL	99.56	99.56	99.50	99.49	99.67	99.61	99.32	99.37
FedProx	98.73	98.90	98.53	98.72	99.02	99.16	98.08	98.31
FedAvg	99.66	99.66	99.61	99.61	99.68	99.72	99.54	99.49
STELLAR	99.55	99.72	99.48	99.68	99.60	99.72	99.37	99.65

performance metrics under IID and Non-IID environments, as shown in Table I.

From Table I, the methods exhibit distinct performance characteristics across environments. In the ideal **IID scenario**, **FedAvg** achieves the highest performance metrics (Accuracy 99.66%), serving as a strong baseline. **STELLAR** follows closely with an accuracy of 99.55%, showing a negligible gap (<0.15%) compared to the full-participation FedAvg. FedProx performs the weakest in both scenarios (~98.7%–98.9%), further confirming that its proximal term limits the model's potential in this specific feature space.

However, the superiority of STELLAR becomes evident in the challenging **Non-IID scenario**. While FedAvg maintains its performance, **STELLAR surpasses all baselines**, achieving the highest Accuracy (99.72%), F1-Score (99.68%), Precision (99.72%), and Recall (99.65%). This indicates that STELLAR's multi-dimensional similarity grouping effectively captures the heterogeneous data distribution, allowing the model to generalize better than random aggregation (FedAvg) or rigid regularization (FedProx).

Most importantly, this performance must be viewed in the context of resource consumption (Fig. 9). Although STELLAR is marginally lower than FedAvg in the IID setting, it achieves this near-optimal performance while consuming **approximately 77% less energy**. In the Non-IID setting, STELLAR achieves a **"double win"**—providing both the highest classification accuracy and the lowest energy cost (80% savings). This fully proves its excellent performance-energy balance for resource-constrained satellite networks.

E. Validation of Asynchronous Aggregation and Staleness Compensation

The main performance comparison (Section IV-D) and the subsequent ablation studies (Sections IV-F–IV-G) assume ideal communication links to isolate algorithmic gains. However, LEO satellite networks face inevitable communication delays due to intermittent orbital visibility and fluctuating onboard capabilities. To validate the asynchronous aggregation protocol proposed in Section III-F, we stress-tested STELLAR under varying delay scenarios on the OneWeb constellation.

1) Experiment Setup: We evaluate three scenarios to test system survivability:

1) **Baseline (Sync)**: 0% delay probability (ideal reference);
2) **Mild Async**: 30% delay, 1–2 rounds lag (typical congestion);
3) **Severe Async**: 60% delay, 1–4 rounds lag (frequent link interruptions).

STELLAR uses staleness compensation with decay coefficient $\lambda = 0.5$ (Section III-F). Traditional synchronous FL either waits indefinitely for stragglers or discards late updates, while STELLAR's time-window protocol salvages them via the staleness weighting mechanism in Eq. 14.

2) Convergence and Stability Analysis: **Robustness Under Severe Delays:** Fig. 10a shows STELLAR maintains convergence even under the *Severe Async* scenario (60% nodes with 1–4 rounds lag). Although minor fluctuations occur (e.g., Round 19 drops to ~98.5%), the model quickly recovers and avoids the divergence or complete stalls typical of strict synchronous protocols. This validates the time-window protocol's effectiveness in harsh space environments.

Mild Asynchrony as Regularization: Interestingly, the *Mild Async* scenario performs comparably to—and occasionally outperforms—the synchronous baseline in early rounds (1)–(5). This suggests moderate asynchronous noise can act as implicit regularization, helping the model escape local optima. However, this benefit diminishes in later stages, confirming that asynchrony is primarily a *robustness mechanism* rather than an optimization technique.

Critical Data Salvaging Capability: Fig. 10b illustrates the mechanism's value in extreme cases. At **Round 12**, zero fresh updates arrived (all ground stations experienced communication failures). Under strict synchronous FL, this would cause *complete training failure* or infinite waiting. However, STELLAR successfully salvaged 4 stale updates from previous rounds, ensuring continuous progress. Over 20 rounds, **49 stale packages** were recovered, providing essential data volume to maintain the convergence trend in Fig. 10a. This demonstrates that the staleness compensation mechanism effectively trades temporal flexibility for maximized data utilization, as theoretically bounded in Lemma 2.

F. Multi-Dimensional Similarity Ablation Study

To quantitatively evaluate the individual contribution of each similarity component and validate the necessity of the proposed multi-dimensional metric design, we conducted an extensive ablation study in the Non-IID scenario.

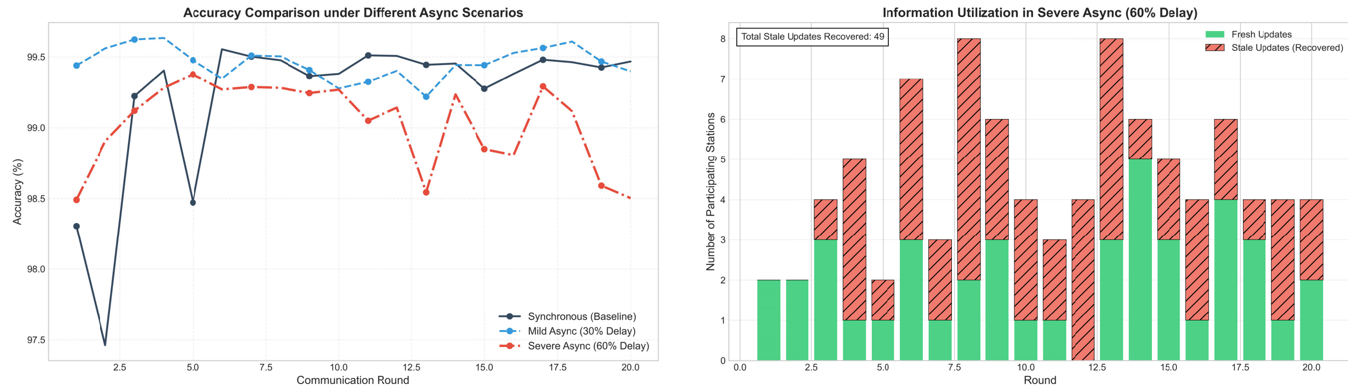
1) Experimental Design: STELLAR's multi-dimensional similarity is a weighted sum of Parameter Similarity (Sim_{param}), Convergence Pattern Similarity (Sim_{loss}), and Prediction Behavior Similarity (Sim_{pred}). We designed two sets of variants for comparison:

- **Single Component Variants**: Only one similarity metric is retained (denoted as *Only-Param*, *Only-Loss*, *Only-Pred*) to observe individual contributions.
- **Component Necessity Variants**: One metric is removed while the other two are retained and equally weighted (denoted as *No-Param*, *No-Loss*, *No-Pred*) to analyze the indispensability of each dimension.

All other experimental conditions (topology, data partition, model structure) remain consistent with previous sections. Each variant was run independently 4 times, and the test accuracy was smoothed via a sliding window to reduce noise.

2) Result Analysis: Fig. 11 presents the ablation results.

Single Component Analysis (Left):



(a) Accuracy under Asynchronous Scenarios (b) Information Utilization (60% Delay)

Fig. 10. Asynchronous aggregation performance. (a) model accuracy remains robust even under severe delays (60% links with 1-4 rounds lag). (b) Stale updates (red hatched bars) are salvaged to maintain data availability—49 total packages recovered over 20 rounds.

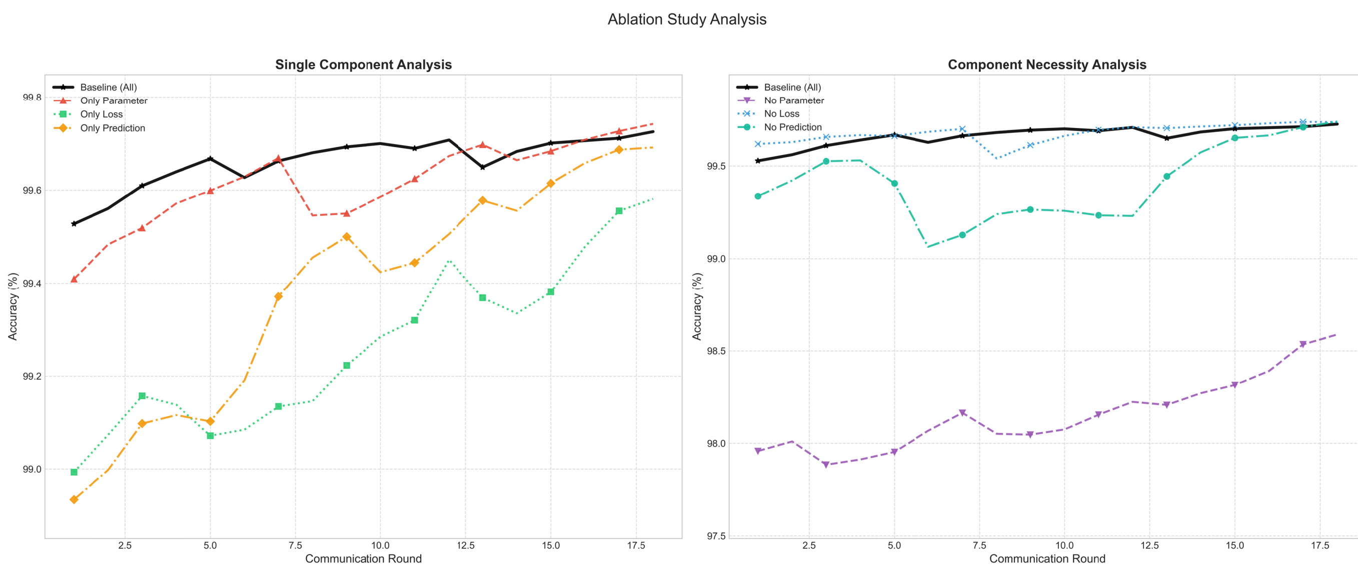


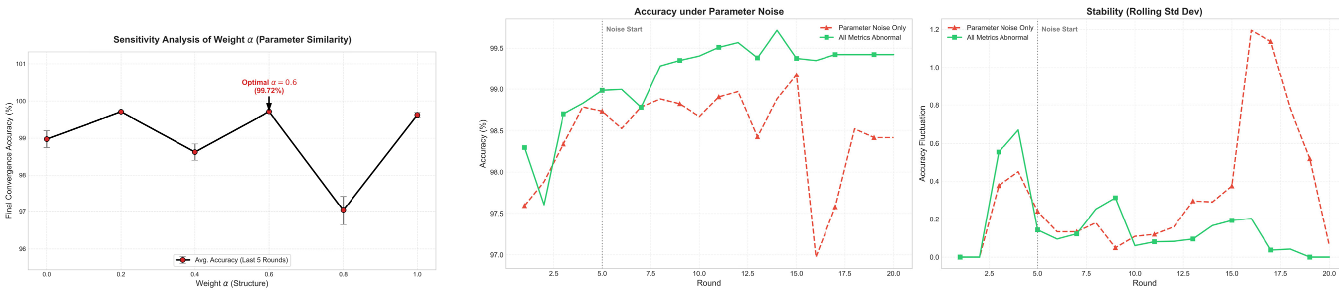
Fig. 11. Ablation study of STELLAR's Multi-dimensional similarity under non-IID Scenario. The left plot compares single-component variants against the full baseline, while the right plot compares variants with one component removed.

- **Superiority of Multi-dimensional Fusion**: The full *Baseline* (Black line) consistently achieves the **highest accuracy and fastest convergence** rate, especially during the first 15 rounds. This empirically proves that fusing multiple dimensions yields better grouping quality than any single metric alone.
- **Dominance of Parameter Similarity**: Among the single-component variants, *Only-Param* (Red line) performs the best and is the closest to the Baseline. However, it still lags behind the Baseline in the early-to-mid training stages, indicating that relying solely on parameters is insufficient for rapid convergence.
- **Auxiliary Role of Behavior Metrics**: Both *Only-Pred* (Orange) and *Only-Loss* (Green) show significantly slower start-up speeds. However, *Only-Pred* catches up in later rounds, suggesting that prediction behavior provides valuable discriminative information as the model matures.

Component Necessity Analysis (Right):

- *No-Param* exhibits the lowest performance, significantly lagging behind the Baseline, confirming that direct parameter similarity is critical for grouping quality.
- *No-Loss* and *No-Pred* perform better than *No-Param* but still show degradation compared to the Baseline. This indicates that convergence patterns and prediction behaviors provide essential supplementary and corrective information.
- All "missing-one" variants show performance degradation, proving that the three dimensions are complementary: simultaneously utilizing parameter, loss, and prediction information yields the most stable and precise grouping.

Conclusion: The ablation study confirms that the three metrics are complementary: Parameter similarity ensures the performance lower bound (Foundation), Prediction behavior guarantees training stability (Stabilizer), and Loss patterns provide fine-grained refinement (Fine-tuner). Simultaneously



(a) Sensitivity (α Analysis) (b) Robustness (Accuracy & Stability)

Fig. 12. Sensitivity and robustness analysis. (a) illustrates the impact of parameter similarity weight α on convergence, identifying α = 0.6 as the optimal configuration. (b) visualizes the system robustness under noise: the left plot shows the accuracy comparison, while the right plot displays the stability (rolling standard deviation). The proposed hybrid metric (α = 0.6) maintains high accuracy and low variance compared to the single-metric approach (α = 1.0).

utilizing all three yields the most stable and precise grouping results.

G. *Sensitivity and Robustness Analysis*

To determine the optimal weight configuration and justify the multi-dimensional design, we conducted a sensitivity analysis followed by a robustness stress test.

1) Experimental Design: First, we analyzed sensitivity by varying α from 0.0 to 1.0 with a step size of 0.2 (setting $\beta = \gamma = \frac{1-\alpha}{2}$). The results (Fig. 12a) identify the hybrid configuration (α = 0.6) as the optimum, though the pure parameter strategy (α = 1.0) remains competitive in ideal settings (gap <0.15%). **To verify whether the multi-dimensional design provides essential stability beyond this marginal accuracy gain**, we designed a stress test comparing the two. Specifically, we injected **Gaussian white noise** into model parameters uploaded by satellites starting from **Round 5** to simulate transmission distortion.

2) Result Analysis: The robustness test (Fig. 12b) confirms our hypothesis. Upon noise injection, the single-metric approach (α = 1.0) becomes highly unstable, with accuracy dropping to ~97.0% and variance spiking. In contrast, the hybrid metric (α = 0.6) demonstrates superior resilience, maintaining accuracy above 99.0% with minimal fluctuations. This proves that auxiliary behavioral metrics effectively offset parameter noise, ensuring system reliability in practical satellite environments.

V. Conclusion

In this paper, we present STELLAR, a similarity-based federated learning framework designed to reconcile the conflict between high detection accuracy and strict resource constraints in LEO satellite networks. By synergizing a multi-dimensional similarity metric with an energy-efficient representative selection mechanism, STELLAR effectively eliminates computational redundancy. Extensive evaluations demonstrate that our framework reduces network-wide energy consumption by 77%–80% while maintaining a near-optimal detection accuracy of 99.72%, even in highly heterogeneous Non-IID environments.

Furthermore, we provided a rigorous theoretical analysis guaranteeing an $O(1/\sqrt{T})$ convergence rate. The proposed time-window asynchronous protocol with staleness compensation ensures system robustness against severe communication delays (up to 60% link instability). Comparative results show that STELLAR consistently outperforms specialized baselines such as FedAvg, FedProx, and SDA-FL. These findings establish STELLAR as a sustainable and robust security paradigm for next-generation mega-constellations where resource optimization is paramount.

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